DEMAND SENSE

An AI-Backed Driver Nudge System for Demand-Aware Repositioning

Ride-hail apps tell drivers WHERE to go. We tell them WHY — and let them decide.

9.4M
YELLOW TAXI TRIPS · JAN–MAR 2023

TEAM 7
Davey Johnson · Hengrui Li · Huiguo Liu · Mansi Malpani · Mounika Polamreddy

THE PROBLEM

Ride-hailing platforms suffer from a critical supply–demand mismatch.

DRIVER IDLE TIME

Drivers wait in the wrong zones, losing earnings while riders queue up across the city.

ZERO VISIBILITY

No reasoning behind dispatch, drivers only see the order to move, never why a zone matters.

LOST REVENUE

Mismatches cause longer waits, cancellations, and fewer completed rides per driver hour.

OUR SOLUTION

A demand-aware driver decision-support system, built on big data infrastructure and grounded LLM reasoning.

SCALE

9.4M

NYC YELLOW TAXI TRIPS

Jan–Mar 2023 · NYC TLC open data joined with the Taxi Zone lookup (263 zones across 5 boroughs).

PySpark · Delta Lake · Unity Catalog

LARGE-SCALE PROCESSING

11-part PySpark pipeline writing five Delta Lake tables.

ZONE DEMAND SCORING

Zone trips ÷ citywide average · low-confidence cells filtered out.

LLM NUDGE GENERATION

8 windows · few-shot prompting · strict grounding rules.

VALIDATED RELIABILITY

Temporal holdout vs a 20% random baseline.

DEMAND INTELLIGENCE PIPELINE

Seven analysis phases, five Delta tables, from raw trip data to validated nudges.

01 INGEST
Load 3 monthly Parquet files + zone lookup

→

02 CLEAN
Dates · zones 1–263 · fare & distance outliers

→

03 AGGREGATE
Zone × day-type × daypart

→

04 SCORE
zone ÷ citywide avg · flag low-confidence

→

05 RANK
Top-5 zones per window × 8 windows

MORNING_RUSH · 6–9:59 | MIDDAY · 10–15:59 | EVENING_RUSH · 16–19:59 | NIGHT · 20–5:59 | × WEEKDAY / WEEKEND

06 NUDGE GENERATION
OpenAI Chat API · few-shot prompt · one grounded message per window.

07 VALIDATION
Temporal holdout hit-rate against real March 2023 demand.

EXPLAINABLE NUDGES

Structured demand signals → LLM → clear, trustworthy driver guidance.

1. **DEMAND RANKING**
   Confidence filter, then Top-5 zones for each of the 8 windows.

2. **FEW-SHOT PROMPTING**
   Gold examples fix tone, length, and grounding before every API call.

3. **NUDGE GENERATED**
   One short, grounded message per window, produced by the Chat API.

4. **QUALITY GUARDRAILS**
   ≤ 60 words · ≤ 2 zones · ~Nx multipliers · no “surge / traffic / earnings.”

★ EXAMPLE OUTPUT

WEEKEND · EVENING RUSH

*“During the weekend evening rush, JFK Airport is about **14.9x** busier than average with roughly 1,200 pickups per evening. East Village follows at about **12.7x** with around 1,000 pickups per evening. **Focus on JFK Airport for potential opportunities.** “*

< 60 words | zones cited | grounded

EVALUATION & RESULTS

Temporal holdout validation, predictions tested against real March demand.

STEP 01
Train on January–February trip patterns

→

STEP 02
Rank top zones per time window

→

STEP 03
Label March zones by demand percentile

→

STEP 04
Compare vs 20% random baseline

HIT RATE · ALL 8 WINDOWS

100%

5× better than random chance (20%)

BEATS 20% BASELINE
Consistent outperformance across the holdout month.

STABLE ACROSS 8 / 8 WINDOWS
Weekday / weekend × four dayparts, no collapse.

8 / 8 NUDGES PASSED
Every generated message cleared all guardrail checks.

KEY IMPACT

What this system delivers for drivers and platforms.

REDUCE IDLE TIME

Guide drivers to high-demand zones before peaks form.

MORE COMPLETED RIDES

Better positioning means more pickups per shift.

DRIVER EARNINGS UP

Data-driven repositioning raises per-hour income.

EXPLAINABLE AI

Drivers know **why** — building trust and higher adoption.

DEMAND SENSE

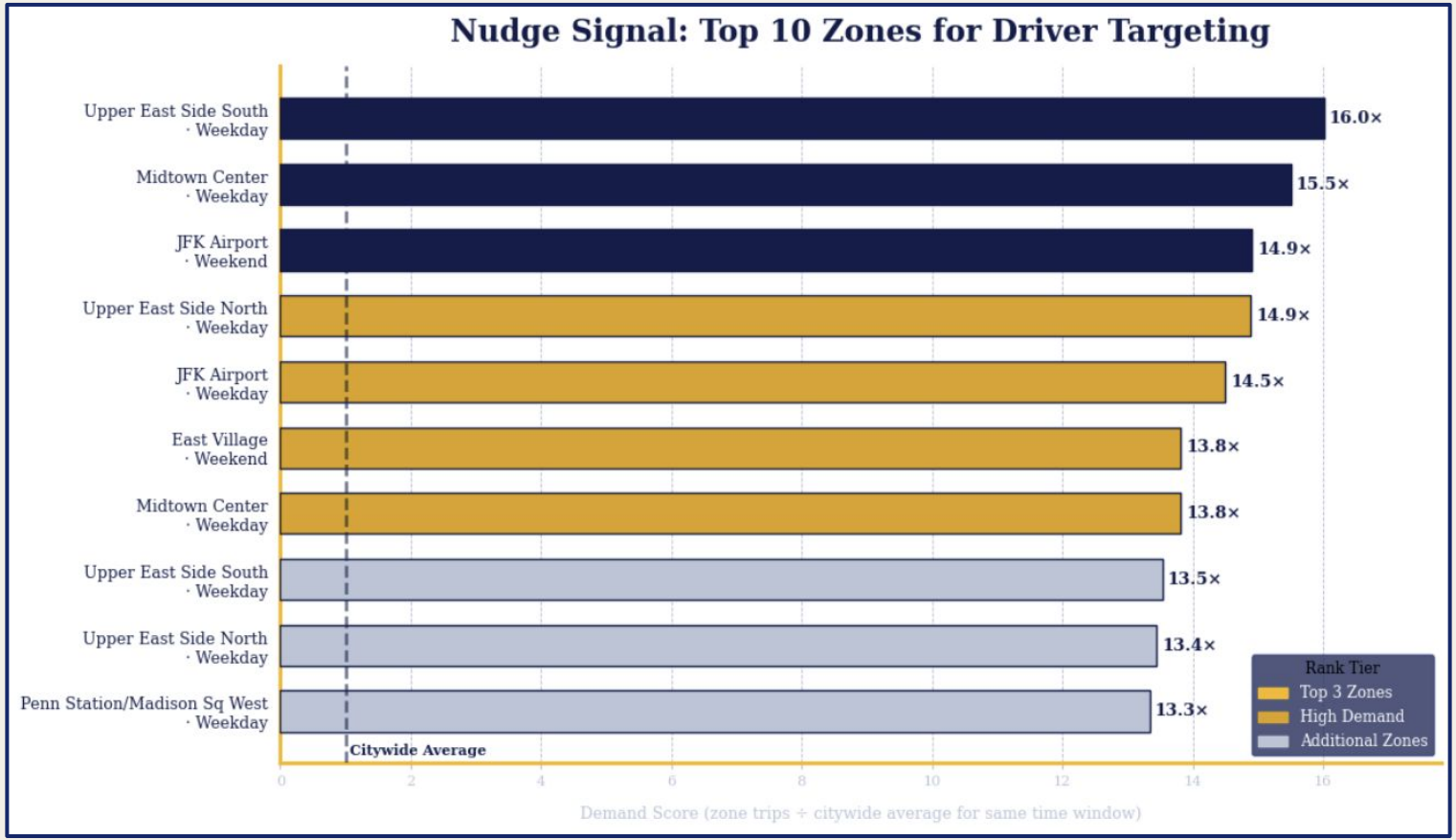
- 9.4M NYC Yellow Taxi trips processed end-to-end
- Normalized zone scoring with confidence filtering
- LLM nudges grounded in real data, not speculation
- **100%** hit rate on March holdout, 5× better than random

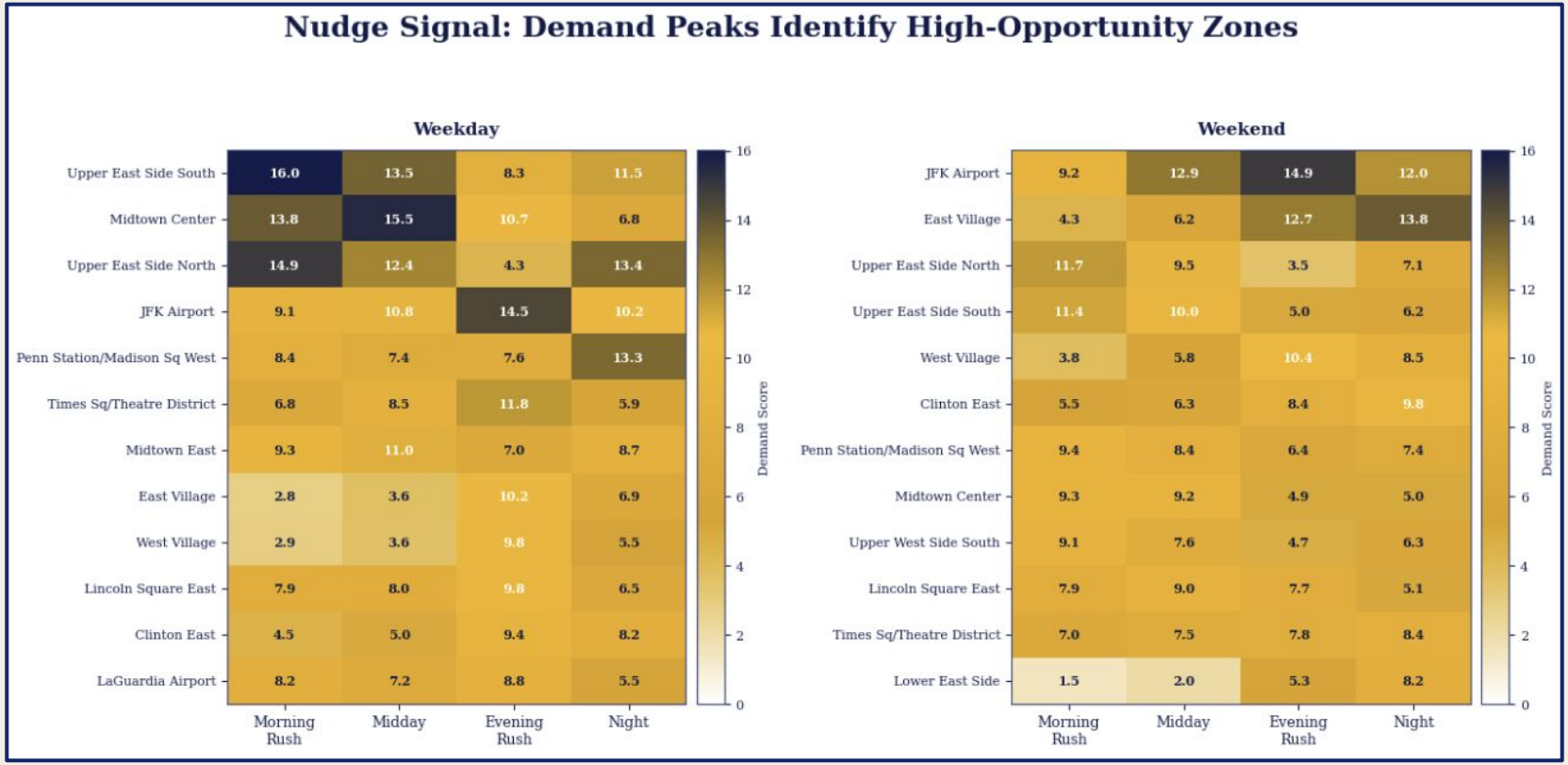
100%

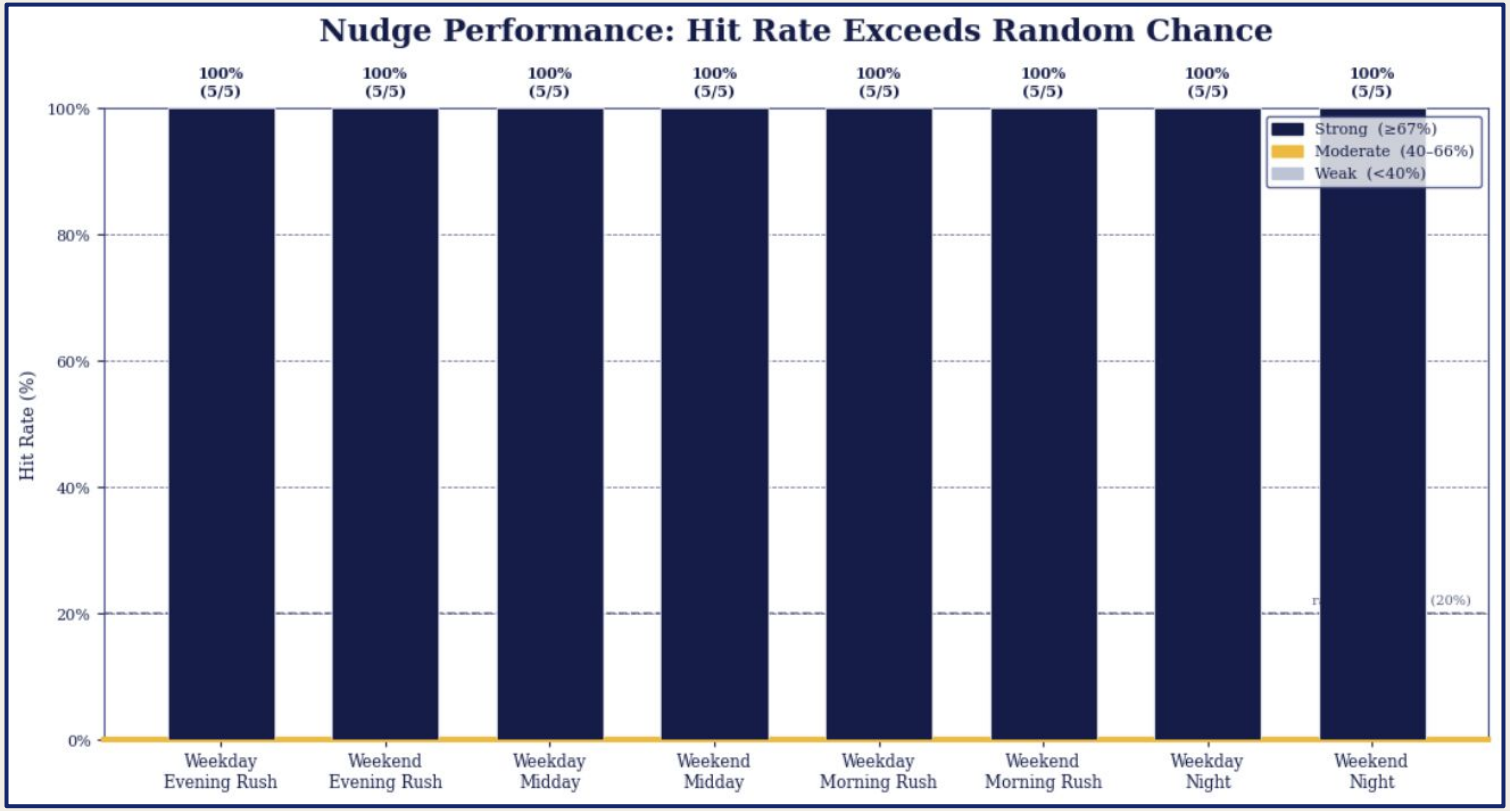
HIT RATE · MARCH HOLDOUT

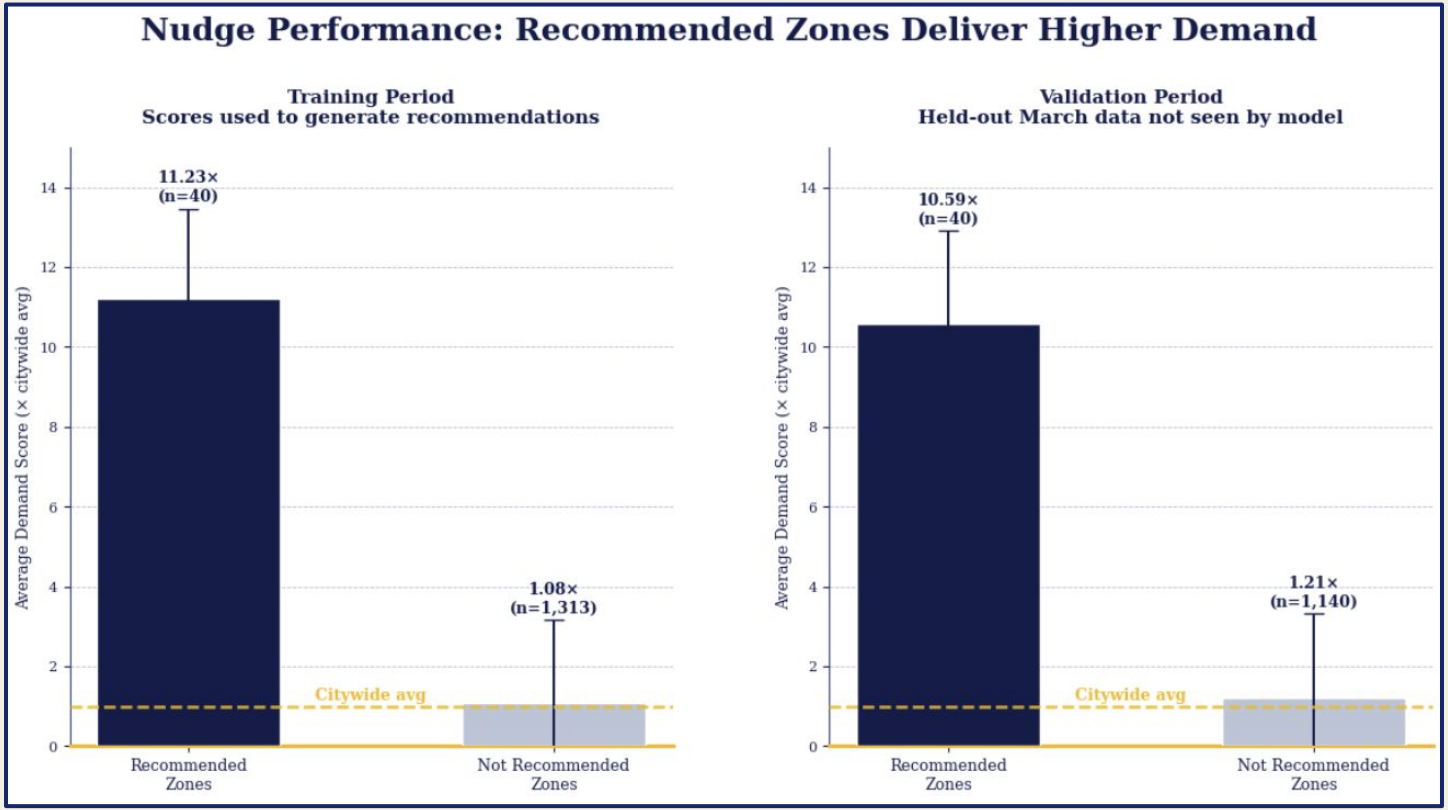
Q&A

Thank you!


Nudge Signal: Top 10 Zones for Driver Targeting
Upper East Side South · Weekday
Midtown Center · Weekday
JFK Airport · Weekend
Upper East Side North · Weekday
JFK Airport · Weekday
East Village · Weekend
Midtown Center · Weekday
Upper East Side South · Weekday
Upper East Side North · Weekday
Penn Station/Madison Sq West · Weekday
16.0×
15.5×
14.9×
14.9×
14.5×
13.8×
13.8×
13.5×
13.4×
13.3×
Citywide Average
0
2
4
6
8
10
12
14
16
Demand Score (zone trips ÷ citywide average for same time window)
Rank Tier
Top 3 Zones
High Demand
Additional Zones


Nudge Signal: Demand Peaks Identify High-Opportunity Zones
Weekday
Upper East Side South 16.0 13.5 8.3 11.5
Midtown Center 13.8 15.5 10.7 6.8
Upper East Side North 14.9 12.4 4.3 13.4
JFK Airport 9.1 10.8 14.5 10.2
Penn Station/Madison Sq West 8.4 7.4 7.6 13.3
Times Sq/Theatre District 6.8 8.5 11.8 5.9
Midtown East 9.3 11.0 7.0 8.7
East Village 2.8 3.6 10.2 6.9
West Village 2.9 3.6 9.8 5.5
Lincoln Square East 7.9 8.0 9.8 6.5
Clinton East 4.5 5.0 9.4 8.2
LaGuardia Airport 8.2 7.2 8.8 5.5
Morning Rush
Midday
Evening Rush
Night
Demand Score
0 2 4 6 8 10 12 14 16
Weekend
JFK Airport 9.2 12.9 14.9 12.0
East Village 4.3 6.2 12.7 13.8
Upper East Side North 11.7 9.5 3.5 7.1
Upper East Side South 11.4 10.0 5.0 6.2
West Village 3.8 5.8 10.4 8.5
Clinton East 5.5 6.3 8.4 9.8
Penn Station/Madison Sq West 9.4 8.4 6.4 7.4
Midtown Center 9.3 9.2 4.9 5.0
Upper West Side South 9.1 7.6 4.7 6.3
Lincoln Square East 7.9 9.0 7.7 5.1
Times Sq/Theatre District 7.0 7.5 7.8 8.4
Lower East Side 1.5 2.0 5.3 8.2
Morning Rush
Midday
Evening Rush
Night
Demand Score
0 2 4 6 8 10 12 14 16


Nudge Performance: Hit Rate Exceeds Random Chance
100% (5/5)
100% (5/5)
100% (5/5)
100% (5/5)
100% (5/5)
100% (5/5)
100% (5/5)
100% (5/5)
Hit Rate (%)
100%
80%
60%
40%
20%
0%
(20%)
Strong (≥67%)
Moderate (40–66%)
Weak (<40%)
Weekday Evening Rush
Weekend Evening Rush
Weekday Midday
Weekend Midday
Weekday Morning Rush
Weekend Morning Rush
Weekday Night
Weekend Night


Nudge Performance: Recommended Zones Deliver Higher Demand
Training Period
Scores used to generate recommendations
Validation Period
Held-out March data not seen by model
Average Demand Score (× citywide avg)
11.23×
(n=40)
1.08×
(n=1,313)
10.59×
(n=40)
1.21×
(n=1,140)
Citywide avg
Recommended Zones
Not Recommended Zones
0
2
4
6
8
10
12
14

Nudge Quality: All Messages Meet Grounding Standards
Grounding Check Results by Nudge Window
WD Evening
WD Midday
WD Morning
WD Night
WE Evening
WE Midday
WE Morning
WE Night
Zone Cited
No Forbidden Words
Under 60 Words
No Raw Score
Nudge Length by Window
Weekday
Weekend
60-word limit
38
37
45
43
40
37
46
37
0
10
20
30
40
50
60
Word Count